Student 1 (15 March 2022)

#	Statement	Answer
1	Geometric distribution is a special case of which distribution	Negative binomial distribution
2	Instead of counting the number of successes we are interested to count the numbers of trails until a success is obtained. Which probability distribution should be used in this case	Binomial distribution
3	Choose the correct formula of mean of multinomial distribution	$E(X_i) = nP_i$
4	The domain 0<x<y<1can be Re-written as	$0 < x < 1 \ , \ x < y < 1$
5	X and Y are independent if and only if	P = 0
6	In normal distribution (sigma square) denotes the	Variance
7	For the standard normal distribution E(z) is equal to which value	E(z)=0
8	Choose the correct number of parameters(s) for normal distribution	μ , σ
9	What is relationship b/w the mean, median and mode of a normal distribution	Are equal
10	In normal distribution, the all odd order moments about mean are	Zero
11	The first Raw moment of the normal distribution is equal to	Expected value or mean
12	The first cumulant k'(0) of normal distribution is	$K_1 = \mu$
13	All the higher cumulant of a normal distribution is	Zero
14	The bivariate normal distribution has	Two independent variable
15	The bivariate distribution can be expressed as	P_{IL}
16	Any subset of a sample space is called	Event
17	The quantile function is called	Percentile function or inverse of cumulative distribution
18	If the distribution satisfies the definition of "random variable" even though it does not Appear random in the everyday sense of the word; it is considered as	Degenerate
19	The expected value of e^{itx} is called	Characteristic function
20	Joint probability distribution of two variable is called	Join problem distribution
21	In-one-to-one transformation, for two old variables, we get	Two linear independent vectors
22	In Which distribution the successive draws are conducted through sampling without replacement	Hyper geometric distribution
23	The variance of the hyper geometric distribution is given by	$\frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$
24	The mean and variance in poisson distribution is given by	Mean and variance are equal

STA642 Objective (Final 2022) **Written By: SYED HUSSNAIN ALI NAQVI**

25	**Which distribution has a unique property that its mean and variance are equal**	**Poisson**
26	**The formula $\frac{a+b}{2}$ *is used to find the mean of which distribution***	**Continues uniform distribution**
27	**For the exponential distribution , if $\theta=1/\lambda$ then θ?**	**Give some value and find $\theta=1/\lambda$**
28	**The given pdf belongs to which distribution (Gamma distribution ka pdf tha)**	**Gamma distribution**
29	**For any fixed value of the scale parameter β as α increase, the shape of the distribution tends to**	**Normality shape**
30	**The limit of probability density function of the β-distribution are**	$0 \le x \le 1$
31	**The shape parameters of beta distribution is**	**α and β**
32	**The 2-parameters of uniform distribution are**	**a and b**
33	**Which two parameters determine the shape of binomial distribution**	**p and n**
34	**What is the formula of finding the variance of binomial distribution**	$\sigma^2 = npq$
35	**For the mth moment about the origin if m=1 then we obtain**	**Mean**
36	**For the mth moment about the origin if m=2 then we obtain**	**Variance**
37	**Consider the equation of circle $(x_1-c_1)^2+(x_2-c_2)^2=a^2$ *which is showing the value of radius***	**a**
38	**How many parameters are in binomial normal distribution**	**$t\omega^{\circ}$, *n and p***
39	**A variable is normally distributed with *N(70,100),what is the mean of this distribution***	**N(70,100)** $70 \Rightarrow mean,\ 100 \Rightarrow \text{var}iance$
40	**In the case of a Bivariate Normal distribution the random variables X and Y are independent If and only if the correlation coefficient ρ is equa to**	**zero**

Student 2 (26 March 2022)

#	Statement	A	B	C	D
1	In one-to-one transformation, for two old variables, we get a	One new variable	Three new variable	Four new variable	Two new variable
2	The shape parameters of beta distribution is	-α and β	α and -β	-α and -β	α and β
3	Discrete probability distribution in which outcome is very small period is classified as a	Poisson distribution	Posterior distribution	Cumulative distribution	Normal distribution
4	The domain 0<x<y<1can be Re-written as	0 < x < 1 , x < y < 1	0 < x+y < 1	0 < y < 1 y < x < 1	0 < x < 1 y < x < 1
5	K(t) is equal to	E(M(t))	Var(M(t))	Log M(t)	M(t)
6	The normal curve is asymptotic to which axis	x-axis	y-axis	z-axis	Both x-axis and y-axis
7	Which two parameter determine the shape of binomial distribution	p and n	p and x	p and q	q and x
8	For a data we find correlation coefficient, now we add a digit 5 in all the data values and then again computed the correlation, then new correlation coefficient is	Will decreases	Will increases by 5%	Will remain shame	Will increase.
9	In which distribution, the numbers of trial are not fixed in advance.	Binomial distribution	Bernoulli distribution	Hyper geometric distribution	Geometric distribution
10	Chose the correct number of normal distribution	1	2	3	4
11	For the standard normal distribution E(z) is equal to which value,	1	0	-1	100
12	The limit of probability density function of the β-distribution are			$0 \le x \le 1$ α,β >0	
13	For the join PMF $P_{y_1y_2}(y_1y_2)$ we say obtain the marginal PMF of y_1 by summing on			Y_2	
14	Any subset of a sample space is called			Event	
15	Generally probability density function is symbolically denoted by			$E_x(x)$	
16	If random variable X is infinite then its expectation			Does not exist	
17	Chose the correct parameters of poisson distribution			λ	
18	The formula $\frac{a+b}{2}$ *is used to find the mean of which*			Continues uniform distribution	

	distribution	
19	The given PDF belongs to which distribution $f(x)=\begin{cases}\frac{1}{\alpha\beta^{\alpha}}x^{\alpha-1}e^{\frac{-x}{\beta}} & 0<x<\infty,\alpha>0,\beta>0\\ 0 & elswhere\end{cases}$	Gamma distribution
20	In which distribution the successive draws are conducted through sampling without replacement	Hyper geometric distribution
21	What is formula of finding the MEAN of binomial distribution	np
22	For any fixed value of the scale parameter β as α increase, the shape of the distribution tends to	Normality shape
23	X and Y are independent if and only if	$F_{xy}(x,y)=f_x(x)\blacklozenge f_y(y)$
24	Two paramenter of uniform distribution are	a and b
25	What is the correct formula of mean of hyper geometric distribution	$\frac{nk}{N}$
26	The experiment continuous until r successive are observed where r is specified in advance. This is an example of which distribution	Negative binomial distribution
27	What is the relationship between mean, median, mode of a normal distribution	Always same
28	How many parameter are in the Bivariate normal distribution	Five
29	The biraviate normal distribution can be expressed as	$N(\mu_x,\mu_y,\sigma_x^2,\sigma_y^2,\rho)$
30	The bivariate normal distribution has	Bell shape
31	In normal distribution, the all odd order moment of about mean are	Zero
32	In general, the expected value of a product is _____ to the product of the expected value that is E(XY)___ E(X)E(Y)	Not equal , $\neq$
33	Which distribution pass the quality that is means and standard deviation are equal	Exponential distribution
34	The row moment of the normal distribution is equal to the	Mean E(X)
35	All the higher commulant of a normal distribution are equal to	Zero
36	In Bivariate Normal Distribution, what is the range limit of variable X and Y	$-\infty < x \ , \ y < \infty$
37	The first commulant $K^{I}(0)$ of normal distribution is	Mean (μ)
38	In the case of Bivariate normal distribution, the random variable X and Y are independent iff the correlation coefficient (ρ) =	$\rho = 0$
39	Choose the correct formula of VARIENCE of multinomial distribution	$np_i(1-p_i)$
40	Consider the equation of circle $(x_1-c_1)^2+(x_2-c_2)^2=a^2$	a

MCQ BINOMIAL AND HYPERGEOMETRIC DISTRIBUTIONS

MCQ 8.1
A Bernoulli trial has:
(a) At least two outcomes (b) At most two outcomes
(c) Two outcomes (d) Fewer than two outcomes

MCQ 8.2
The two mutually exclusive outcomes in a Bernoulli trial are usually called:
(a) Success and failure (b) Variable and constant
(c) Mean and variance (d) With and without replacement

MCQ 8.3
Nature of the binomial random variable X is:
(a) Quantitative (b) Qualitative **(c) Discrete** (d) Continuous

MCQ 8.4
In a binomial probability distribution, the sum of probability of failure and probability of success is always:
(a) Zero (b) Less than 0.5 (c) Greater than 0.5 **(d) One**

MCQ 8.5
Ina binomial experiment, the successive trials are:
(a) Dependent **(b) Independent** (c) Mutually exclusive (d) Fixed

MCQ 8.6
The parameters of the binomial distribution are:
(a) n and p (b) p and q (c) np and nq (d) np and npq

MCQ 8.7
The range of binomial distribution is:
(a) 0 to n (b) 0 to ∞ (c) -1 to +1 (d) 0 to 1

MCQ 8.8
The mean and standard deviation of the binomial probability distribution 'are respectively:
(a) np and npq **(b) np and $\sqrt{npq}$** (c) np and nq (d) n and p

MCQ 8.9
In a binomial experiment with three trials, the variable can take:
(a) 2 values (b) 3 values **(c) 4 values** (d) 5 values

MCQ 8.10
The shape of the binomial probability distribution depends upon the values of its:
(a) Mean (b) Variance **(c) Parameters** (d) Quartiles

MCQ 8.11
In binomial distribution the numbers of trials are:
(a) Very large (b) Very small **(c) Fixed** (d) Not fixed

MCQ 8.12
In a binomial probability distribution, relation between mean and variance is:
(a) Mean < Variance (b) Mean = Variance
(c) Mean > Variance (d) Difficult to tell

MCQ 8.13
In binomial distribution when n = 1, then it becomes:
(a) Hypergeometric distribution (b) Normal distribution
(c) Uniform distribution **(d) Bernoulli distribution**

MCQ 8.14
The mean of a binomial distribution depends on:
(a) Number of trials (b) Probability of success
(c) Probability of failure **(d) Number of trials and probability of success**

MCQ 8.15
The variance of a binomial distribution depends on:
(a) Number of trials (b) Probability of success
(c) Probability of failure **(d) All of the above**

MCQ 8.16
Which of the following is not property of a binomial experiment?
(a) Probability of success remains constant
(b) **n** is fixed
(c) Successive trials are dependent
(d) It has two parameters

MCQ 8.17
The binomial probability distribution is symmetrical when:
(a) p = q (b) $p < q$ (c) $p > q$ (d) $np > npq$

MCQ 8.18
The binomial distribution is negatively skewed if:
(a) $p < 1/2$ (b) $p = 1/2$ **(c) $p > 1/2$** (d) $p = 1$

MCQ 8.19
In a binomial probability distribution, the skewness is positive for:
(a) $p < 1/2$ (b) $p = 1/4$ (c) $np = npq$ (d) $np = nq$

MCQ 8.20
Which of the following statements is false?
(a) Expected value of a constant
(b) In a binomial distribution the standard deviation is always less than its variance
(c) In a binomial distribution the mean is always greater than its variance
(d) In binomial experiment the probability of success remains constant from trial to trial

MCQ 8.21
If a binomial probability distribution has parameters (n, p)= (5, 0.6), the probability of x = 3.5 is:
(a) 0 (b) 1 (c) 0.6 (d) 0.4

MCQ 8.22
In a binomial experiment n= 4, P(x=2) = 216/625 and P(x=3) = 216/625. P(x=-2) is:
(a) 216/625 (b) 1 **(c) 0.6** (d) Difficult to tell

MCQ 8.23
If n = 6 and p= 0.9 then the value of P(x=7) is:
(a) Zero (b) Less than zero (c) More than zero (d) One

MCQ 8.24
In a binomial probability distribution, coefficient of skewness = $(q - p)/\sqrt{npq} = 0$, it means that the distribution is:
(a) Symmetrical (b) Skewed to the left (c) Skewed to the right (d) Highly skewed

MCQ 8.25
For a binomial distribution with n = 10, p = 0.5, the probability of zero or more successes is:
(a) 1 (b) 0.5 (c) 0.25 (d) 0.75

MCQ 8.26
In a binomial distribution, the mean, median and mode coincide when:
(a) $p < 1/2$ (b) $p > ½$ (c) $p \neq 1/2$ **(d) p = 1/2**

MCQ 8.27
In which distribution, the probability success remains constant from trial to trial?
(a) Hypergeometric distribution **(b) Binomial distribution**
(c) Sampling distribution (d) Frequency distribution

MCQ 8.28
In a binomial experiment when n = 5, the maximum number of successes will be:
(a) 0 (b) 2.5 (c) 4 **(d) 5**

MCQ 8.29
In a binomial experiment when n = 10, the minimum number of successes will be:
(a) 0 (b) 5 (c) 10 (d) 11

MCQ 8.30
If n = 10 and p = 0.6, then $P(x \geq 0)$ is:
(a) 0.5 (b) 0.6 **(c) 1.0** (d) 1.2

MCQ 8.31
A random variable X has a binomial distribution with n = 4, the standard deviation of X is:
(a) 4 pq **(b) $2\sqrt{pq}$** (c) 4 p (d) 4 (q+p)

MCQ 8.32
In a multiple choice test there are five possible answers to each of 20 questions. If a candidate guesses the correct answer each time, the mean number of correct answers is:
(a) 4 (b) 5 (c) 1/5 (d) 20

MCQ 8.33
If three coins are tossed, the probability of two heads is:
(a) 1/8 **(b) 3/8** (c) 2/3 (d) 0

MCQ 8.34
Random variable x has binomial distribution with n = 8 and p = ½.. The most probable value of X is:
(a) 2 (b) 3 **(c) 4** (d) 5

MCQ 8.35
The value of second moment about the mean in a binomial distribution is 36. The value of the standard deviation of a binomial distribution is:
(a) 36 **(b) 6** (c) 1/36 (d) 1/6

MCQ 8.36
For a binomial probability distribution, the expected frequency of x successes in N experiments is:
(a) $\frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$ (b) $p^x(1-p)^{1-x}$
(c) $N\binom{n}{x}p^xq^{n-x}$ (d) $\binom{n}{x}p^x(1-p)^{n-x}$

MCQ 8.37
In a binomial frequency distribution $100\,(1/5 + 4/5)^5$. The parameters n and p are respectively:
(a) (5, 1/5) (b) (1/5, 4/5) (c) (100, 4/5) **(d) (5, 4/5)**

MCQ 8.38
For a binomial frequency distribution $100\,(1/5 + 4/5)^5$, the mean is:
(a) 1/5 (b) 4/5 (c) 5 **(d) 4**

MCQ 8.39
For a binomial distribution (1/3 + 2/3)18, the standard deviation of the binomial distribution will be:
(a) 2 (b) 4 (c) 6 (d) 12

MCQ 8.40
The hypergeometric distribution has:
(a) One parameter (b) Two parameters **(c) Three parameters** (d) Four parameters

MCQ 8.41
The parameters of the hypergeometric distribution are:
(a) N, n, p (b) N, n, np **(c) N, n, k** (d) n and p

MCQ 8.42
Nature of the Hypergeometric random variable is:
(a) Continuous **(b) Discrete** (c) Qualitative (d) Quantitative

MCQ 8.43
In hypergeometric· distribution, the successive trials are:
(a) Independent **(b) Dependent** (c) Very large (d) Very small

MCQ 8.44
In a hypergeometric distribution, the probability of success:
(a) Remains constant from trial to trial
(b) Does not remain constant from trial to trial
(c) Equal to probability of failure
(d) Less than probability of failure

MCQ 8.45
If in a hypergeometric distribution N = 10, k = 5 and n = 4; then the probability of failure is:
(a) 2 **(b) 0.5** (c) 1 (d) 0.25

MCQ 8.46
The rang of hypergeometric distribution is:
(a) 0 to n (b) 0 to k (c) 0 to N **(d) 0 to n or k (whichever is less)**

MCQ 8.47
The number of trials in hypergeometric distribution is:
(a) Not fixed **(b) Fixed** (c) Large (d) Small

MCQ 8.48
The probability of a success changes from trial to trial in:
(a) Binomial distribution
(b) Hypergeometric distribution
(c) Normal distribution
(d) Frequency distribution

MCQ 8.49
The mean of the hypergeometric distribution is:
$(a)\frac{nk}{N}$ $(b)\frac{Nk}{n}$ $(c)\frac{Nn}{k}$ $(d)\frac{n+k}{N}$

MCQ 8.50
The standard deviation of the hypergeometric distribution is:
$(a)\frac{nk(k-N)(N-n)}{N^2(N-1)}$
$(b)\sqrt{\frac{nk(N-k)(N-n)}{N^2(N-1)}}$
$(c)\sqrt{\frac{nk(N-k)(n-N)}{N^2(N-1)}}$
$(d)\sqrt{\frac{nk(N-k)+(N-n)}{N^2(N-1)}}$

MCQ 8.51
In hypergeometric probability distribution, the relation between mean and variance is:
(a) Mean > variance (b) Mean < Variance (c) Mean = Variance (d) Mean = 2Variance

MCQ 8.52
Which of the following is the property of hypergeometric experiment?
(a) **p** remains constant from trial to trial
(b) Successive trials are independent
(c) Sampling is performed without replacement
(d) **n** is not fixed

MCQ 8.53
Hypergeometric distribution reduces to binomial distribution when:
(a) N = n (b) $n \to \infty$ **(c) $N \to \infty$** (d) N < n

MCQ 8.54
In a hypergeometric distribution N=6, n=4 and k=3, then the mean is equal to:
(a) 2 (b) 4 (c) 6 (d) 24

MCQ 8.55
Given N = 11, n = 5, k = 7; P(x ≥ 1) equals:
(a) 1 (b) 1/66 (c) 65/66 (d) None of the above

MCQ 8.56
Given N =12, n =5, k= 4; P(x ≤ 4) equals:
(a) Less than one **(b) Exactly one** (c) More than one (d) Between 0.5 and 1

1.(c)	2.(a)	3.(c)	4.(d)	5.(b)	6.(a)	7.(a)	8.(b)	9.(c)	10.(c)	11.(c)	12.(c)	13.(d)	14.(d)	15.(d)
16.(c)	17.(a)	18.(c)	19.(a)	20.(b)	21.(a)	22.(c)	23.(a)	24.(a)	25.(a)	26.(d)	27.(b)	28.(d)	29.(a)	30.(c)
31.(b)	32.(a)	33.(b)	34.(c)	35.(b)	36.(c)	37.(d)	38.(d)	39.(a)	40.(c)	41.(c)	42.(b)	43.(b)	44.(b)	45.(b)
46.(d)	47.(b)	48.(b)	49.(a)	50.(b)	51.(a)	52.(c)	53.(c)	54.(a)	55.(a)	56.(b)				